1. Early Systems of Classification

Aristotle 350 BC	Two Kingdom (Linnaeus 1753)	Limitations

- Living organisms have been classified since ancient times, but early classification was **not based on scientific criteria** — it relied on simple, visible features.
- **Aristotle** (350 BC) made the earliest notable attempt, using gross morphology:
- Classified **plants** into *trees, shrubs,* and *herbs* based on size and structure.
- Classified **animals** by presence of red blood: **Enaima** (red blood) vs **Anaima** (no red blood).
- Also grouped animals by habitat into **Aquatic** and **Terrestrial** forms.

Linnaeus and the Two Kingdom System

- **Carolus Linnaeus (1753)** proposed the **Two Kingdom system**, dividing all living organisms into **Kingdom Plantae** and **Kingdom Animalia**.
- This system used easily observable features — mode of nutrition and cell wall presence — as the main dividing line between plants and animals.

Why the Two Kingdom System Failed

- It did **not distinguish eukaryotes from prokaryotes** — bacteria (prokaryotic) were grouped with plants (eukaryotic).
- It did **not distinguish unicellular from multicellular** organisms — e.g. unicellular *Chlamydomonas* and multicellular *Spirogyra* were both placed under 'algae'.
- It did **not distinguish photosynthetic from non-photosynthetic** organisms — green algae (autotrophic) and fungi (heterotrophic) were both grouped as 'plants'.
- It grouped **prokaryotic bacteria and blue-green algae** together with eukaryotic plants, ignoring the fundamental difference in cell structure.
- It did not separate **heterotrophic fungi** (cell wall made of **chitin**) from **autotrophic plants** (cell wall made of **cellulose**).
- **Exam tip:** the Two Kingdom system considered only *gross morphology* — it completely ignored cell structure, mode of nutrition, habitat, reproduction, and evolutionary relationships.

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2. Five Kingdom Classification

Whittaker 1969	4 criteria	Monera > Protista > Fungi > Plantae > Animalia

- Proposed by **R.H. Whittaker (1969)** — the most advanced and widely accepted classification system.

- **Criteria used**, in order of application: Cell structure → Thallus organisation → Mode of nutrition → Mode of reproduction → Phylogenetic (evolutionary) relationships.
- This resulted in **five kingdoms**: **Monera, Protista, Fungi, Plantae,** and **Animalia**.

Character	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Non-cellulosic (peptidoglycan)	Present in some forms	Chitin (no cellulose)	Cellulose	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular / loose tissue	Tissue / organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic + Heterotrophic	Autotrophic + Heterotrophic	Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic)	Heterotrophic

Merits of the Five Kingdom System

- All prokaryotic organisms are neatly grouped together under **Kingdom Monera**.
- Unicellular eukaryotes previously split between 'plants' and 'animals' — such as *Chlamydomonas, Chlorella, Paramecium,* and *Amoeba* — are now united under **Kingdom Protista**.
- Fungi are correctly separated from plants based on their chitin cell wall and heterotrophic nutrition.
- The system reflects evolutionary (phylogenetic) relationships far better than earlier systems.

3. Kingdom Monera

Bacteria only	Archaebacteria	Eubacteria	Mycoplasma

- Bacteria are the **sole members** of Kingdom Monera; they are the most abundant micro-organisms on Earth.
- Found in almost every habitat: hot springs, deep oceans, snow, deserts; many live as parasites inside other organisms.
- Flagella, when present, are made of the protein **flagellin**; the cell wall, when present, is made of **peptidoglycan**; there are no membrane-bound organelles.
- **Shape-based groups:** *Coccus* (spherical), *Bacillus* (rod-shaped), *Vibrio* (comma-shaped), *Spirillum* (spiral).
- **Modes of nutrition:** Autotrophic — Chemosynthetic (uses inorganic substrates) or Photosynthetic (uses CO_2 + H_2O); Heterotrophic — Parasitic or Saprophytic.

A. Archaebacteria (the 'living fossils')

- Live in some of the most extreme and harsh habitats on Earth; mostly **anaerobic**.
- **Halophiles** — thrive in extremely salty areas.
- **Thermoacidophiles** — thrive in hot, acidic springs.
- **Methanogens** — found in marshy areas and in the guts of ruminant animals, where they produce **methane (biogas)** from dung.
- Their unique cell wall structure is what allows them to survive such extreme conditions.

B. Eubacteria (the 'true bacteria')

- Possess a rigid cell wall; have a flagellum if they are motile.
- **Autotrophic Eubacteria:**
- • Photosynthetic — Cyanobacteria (blue-green algae): contain chlorophyll *a*; may be unicellular, colonial, or filamentous; possess a gelatinous sheath; fix atmospheric nitrogen in specialised cells called **heterocysts** (e.g. *Nostoc, Anabaena*).
- • Chemosynthetic — oxidise substances like nitrites, nitrates, and ammonia; important in nutrient recycling.
- **Heterotrophic Eubacteria (decomposers):** the most abundant bacterial group; help make curd from milk, produce antibiotics, and fix nitrogen in the root nodules of legumes.
- **Important pathogens:** cholera (*Vibrio cholerae*), typhoid (*Salmonella typhi*), citrus canker (*Xanthomonas axonopodis*), anthrax (*Bacillus anthracis*).

Mycoplasma

- The **smallest known living organism**.
- **Completely lacks a cell wall** — unique among all bacteria.
- Can survive without oxygen; many species are pathogenic.

Reproduction in Bacteria

- **Asexual** — mainly by **fission** (binary splitting).
- Under unfavourable conditions, bacteria form resistant **spores** to survive.
- **Sexual-like reproduction** — through **conjugation**, where DNA is transferred from one bacterium to another.

4. Kingdom Protista

Unicellular eukaryotes	5 classes	Chryso > Dino > Euglena > Slime > Protozoa

- All members are **unicellular eukaryotes**; mostly found in aquatic habitats.
- Possess a well-defined nucleus and membrane-bound organelles; flagella/cilia are made of the protein **tubulin**.
- Reproduce both asexually and sexually (through cell fusion and zygote formation).

1. Chrysophytes (planktons)

- Microscopic organisms that float passively in fresh water and marine habitats; mostly photosynthetic.
- Includes **diatoms** and golden algae (desmids).
- **Diatoms** have a cell wall made of two thin, overlapping shells embedded with **silica**, making them indestructible.
- Fossilised diatom deposits form **'diatomaceous earth'**, used in polishing, filtration, and sound/fire-proofing.
- Diatoms are considered the **chief 'producers'** of the ocean.

2. Dinoflagellates

- Mostly marine and photosynthetic; coloured yellow, green, brown, blue, or red depending on their pigments.
- Cell wall made of stiff cellulose plates; possess two flagella (one longitudinal, one transverse) lying in a groove.
- **Red tides** occur when *Gonyaulax* multiplies rapidly, turning the sea red and releasing toxins that can kill marine animals.

3. Euglenoids

- Mostly found in fresh, stagnant water.
- Instead of a cell wall, they have a protein-rich, flexible layer called the **pellicle**.
- Possess two flagella of unequal length (one short, one long); pigments similar to those of higher plants.
- **Mixotrophic:** behave as autotrophs in sunlight and heterotrophs (predatory) in the dark — making them a **connecting link between plants and animals**.

4. Slime Moulds

- Saprophytic protists that move slowly over decaying twigs and leaves.
- Under favourable conditions, they form an aggregated mass called the **plasmodium**.

- Under unfavourable conditions, they form fruiting bodies bearing spores.
- Spores have true, resistant walls and are dispersed by air currents.

5. Protozoans (all heterotrophic; primitive relatives of animals)

Type	Key Feature	Example
Amoeboid	Move using pseudopodia; some marine forms have silica shells	Amoeba; Entamoeba (amoebic dysentery)
Flagellated	Free-living or parasitic	Trypanosoma (sleeping sickness); Leishmania (kala-azar)
Ciliated	Thousands of cilia; possess a gullet cavity	Paramoecium
Sporozoans	Have a spore-like infectious stage	Plasmodium (malaria)

5. Kingdom Fungi

Hyphae / Mycelium | **Chitin cell wall** | **Phyco > Asco > Basidio > Deutero**

- Heterotrophic organisms; most are saprophytic, some are parasitic; show huge diversity in morphology and habitat.
- Except **yeast** (unicellular), fungi are multicellular and filamentous — made up of thread-like **hyphae** that together form a network called the **mycelium**.
- **Coenocytic hyphae** are continuous and multinucleate (lack cross-walls); others have **septate hyphae** with cross-walls (septa).
- Cell wall is made of **chitin** and other polysaccharides.
- Symbiotic associations: with algae → **lichens**; with roots of higher plants → **mycorrhiza**.

Reproduction in Fungi

- **Vegetative** — by fragmentation, fission, or budding.
- **Asexual** — through spores such as conidia, sporangiospores, or zoospores.
- **Sexual** — through oospores, ascospores, or basidiospores; involves three steps:
- 1. **Plasmogamy** — fusion of protoplasm of two cells.
- 2. **Karyogamy** — fusion of the two nuclei.
- 3. **Meiosis in the zygote** — produces haploid spores.
- Fusion of compatible haploid hyphae may form a diploid (2n) cell directly, or pass through an intervening **dikaryotic (n+n) stage** called the **dikaryophase** — seen mainly in **Ascomycetes** and **Basidiomycetes**.

Classes of Fungi

Class	Mycelium	Key Feature	Examples
Phycomycetes (algal fungi)	Aseptate, coenocytic	Aquatic/damp habitats; motile zoospores or aplanospores; zygospores formed (isogamous, anisogamous, or oogamous)	Mucor, Rhizopus (bread mould), Albugo
Ascomycetes (sac fungi)	Branched, septate	Conidia produced exogenously on conidiophores; ascospores formed inside sac-like asci within ascocarps	Aspergillus, Claviceps, Penicillium, Neurospora
Basidiomycetes (club fungi)	Branched, septate	No asexual spores; plasmogamy by fusion of vegetative cells forms a dikaryon; basidium then produces 4 exogenous basidiospores in basidiocarps	Agaricus (mushroom), Ustilago (smut), Puccinia (rust)
Deuteromycetes (imperfect fungi)	Septate, branched	Sexual phase absent/unknown; reproduce only through conidia; reclassified once sexual phase is found; important in mineral cycling	Alternaria, Colletotrichum, Trichoderma

Exam-Trap Disease Notes

- **Penicillin**, the 'wonder drug', is extracted from *Penicillium notatum* and *P. chrysogenum*.
- *Alternaria solani* causes **early blight of potato**; *Phytophthora infestans* causes **late blight of potato**.
- *Colletotrichum* causes **red rot of sugarcane**.

6. Kingdom Plantae & Kingdom Animalia

Alternation of generation	5 plant divisions	Animalia overview

Kingdom Plantae

- All members are eukaryotic and photosynthetic; a few show partial heterotrophy:
- • **Insectivorous:** *Dionaea* (Venus flytrap), Bladderwort, *Nepenthes* (pitcher plant), *Sarracenia*, *Drosera* (sundew).
- • **Parasitic:** *Cuscuta*.
- Cells are eukaryotic with prominent chloroplasts and a **cellulosic cell wall**.
- Plants exhibit **alternation of generation** — the diploid sporophytic phase alternates with the haploid gametophytic phase.

- **Five major divisions:** Algae, Bryophyta, Pteridophyta, Gymnosperms, and Angiosperms.

Kingdom Animalia

- Heterotrophic, eukaryotic, and multicellular; **cells lack a cell wall**.
- Depend directly or indirectly on plants for food; digestion is internal; food is stored as **glycogen or fat**.
- Show a definite growth pattern towards an adult shape and size.
- Higher forms show elaborate sensory and neuromotor mechanisms; most are capable of locomotion.
- Sexual reproduction occurs through copulation of male and female, followed by embryological development.

7. Viruses, Viroids & Prions

Acellular	Ivanowsky > Beijerinck > Stanley	Viroids (Diener)	Prions

Viruses

- **Acellular** in nature — not placed under Whittaker's five kingdoms.
- Not considered 'truly living' — they are inert, crystalline particles outside a host, showing life-like activity only inside a living cell.
- **Obligate parasites**; they hijack the host cell's machinery to replicate, often killing the host in the process.
- The term 'virus' (meaning venom or poisonous fluid) was coined by **Pasteur**.
- **D.J. Ivanowsky (1892)** discovered agents smaller than bacteria — able to pass through bacteria-proof filters — causing tobacco mosaic disease.
- **M.W. Beijerinck (1898)** showed that extract from an infected plant could cause disease in a healthy plant, and named it *Contagium vivum fluidum*.
- **W.M. Stanley (1935)** crystallised viruses and showed that the crystals were largely composed of protein.

Structure

- Genetic material is **either DNA or RNA, never both** — forming a nucleoprotein that is infectious.
- **Plant viruses** (e.g. TMV) typically contain single-stranded RNA; **Bacteriophages** contain double-stranded DNA; **Animal viruses** may contain ssRNA, dsRNA, or dsDNA.
- The protein coat is called the **capsid**, made of subunits called **capsomeres**, arranged helically or polyhedrally.
- Human diseases: mumps, smallpox, herpes, influenza, AIDS. Plant diseases: mosaic patterns, leaf rolling/curling, yellowing, vein clearing, dwarfing, stunting.

- A **virion** is a completely assembled virus particle found outside the host.

Viroids

- Discovered by **T.O. Diener (1971)**; smaller and simpler than viruses.
- **Lack a protein coat** — exist as free, low molecular weight RNA.
- Cause diseases such as **potato spindle tuber disease** (PSTVD, transmitted by aphids) and **chrysanthemum stunt disease**.

Prions

- **Infectious, abnormally-folded proteins**, similar in size to viruses; contain no nucleic acid at all.
- Cause neurological diseases: **BSE** (mad cow disease) in cattle, and its human analogue, **CJD** (Creutzfeldt-Jakob disease).

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8. Lichens (Symbiotic Association)

Fungi + Algae	Phycobiont / Mycobiont	Pollution indicator

- A **lichen** is a **symbiotic association** between a fungus and an alga.
- **Phycobiont** — the algal partner (autotrophic); may be a green alga or a cyanobacterium.
- **Mycobiont** — the fungal partner (heterotrophic); typically drawn from Ascomycetes or Basidiomycetes.
- The alga prepares food for the fungus through photosynthesis; the fungus absorbs water and minerals and provides shelter and protection to the alga.
- Lichens are **excellent pollution indicators** — they fail to grow in polluted areas, making their presence (or absence) ecologically significant.